Activity - 2

1) A CCTV image contains Salt-and-Pepper noise due to transmission errors.

= Noise : Salt-and-Pepper noise (Impulse Noise) caused by transmission errors.

Filter : Median Filter used to remove Salt-and-Pepper noise while preserving the edge of the image.

2) A Satellite image is corrupted by Gaussian noise from sensors.

Noise : Gaussian Noise caused by sensor noise in the Satellite image.

Filter : Gaussian Filter (or mean filter) is used to reduce Gaussian noise and smooth the image

3) A medical X-ray image has random noise but fine edges must be preserved.

Noise : Random Noise

Filter :- Bilateral Filter is used to remove random noise while preserving the fine edges and important details in X-ray image.

4) A Scanned document has black and white dots scattered randomly

Noise : Salt-and-Pepper Noise (black and white dots)

Filter: Median Filter is used to remove black and white dots while preserving the text and edges in the scanned document.

5) A low-light image shows heavy grain without clear boundaries.

Noise :- Gaussian Noise (grainy noise due to low-light conditions)

Filter: Gaussian Filter is used to reduce the grainy Gaussian noise and produce a smoother image.

6) Foggy road image has very low contrast

Problem: A foggy road image has very low contrast

Techniques: Histogram Equalization

7) Analyze underwater image appears dull and washed out.

Problem: The underwater image appears dull and washed out.

Technique: Contrast Stretching

8) A chest x-ray shows poor visibility of internal structure

Problem: A chest x-ray show poor visibility of internal structure

Technique: Histogram Equalization

9) A night time traffic image lacks intensity

Problem: The image lacks intensity variation

Technique: Histogram Equalization

10) Grayscale image use only a narrow intensity range.

Problem: The grayscale image use narrow intensity image

Technique: Contrast Stretching